BASH SCRIPTING ASSIGNMENTS

Exercise 1: Create a Script that Greets the User

Write a bash script that:

- Prompts the user for their name.
- Outputs a greeting that includes their name.

Exercise 2: Check if a File Exists

Write a bash script that:

- Prompts the user to enter a filename.
- Checks if the file exists in the current directory.
- Displays an appropriate message indicating whether the file was found or not.

Exercise 3: Display the Current Date and Time

Write a bash script that:

- Prints the current date and time in a readable format.
- Displays a message like "The current date and time is: [date/time]".

Exercise 4: Simple Calculator

Write a bash script that:

- Prompts the user to enter two numbers.
- Asks for an operation (add, subtract, multiply, divide).
- Performs the operation and displays the result.

Exercise 5: Count Files in a Directory

Write a bash script that:

- Prompts the user to enter a directory path.
- Counts the number of files (not directories) inside it.
- Displays the count to the user.

Exercise 6: Check if a User is Logged In

Write a bash script that:

- Prompts the user to enter a username.
- Checks if that user is currently logged into the system.
- Displays a message saying whether the user is logged in or not.

Exercise 7: Number Comparison

Write a bash script that:

- Prompts the user to enter two numbers.
- Compares the numbers and displays which one is greater, or if they are equal.

Exercise 8: Loop Through Numbers

Write a bash script that:

- Prompts the user to enter a number `N`.
- Uses a `for` loop to print all numbers from 1 to `N`.
- Displays "Loop completed!" at the end.

Exercise 9: Display System Information

Write a bash script that:

- Displays basic system information such as hostname, kernel version, and uptime.

Exercise 10: Backup a File

Write a bash script that:

- Prompts the user for a filename.
- Creates a backup of the file with a `.bak` extension.
- Displays a message confirming the backup.

Exercise 11: Check Internet Connectivity

Write a bash script that:

- Pings `google.com` or another reliable host.
- Displays whether the system is connected to the internet.

Exercise 12: Display Disk Usage

Write a bash script that:

- Displays the disk usage of the current directory using the `du` command.
- Shows the total size in human-readable format.

Exercise 13: Print Even Numbers

Write a bash script that:

- Prompts the user to enter a number `N`.
- Prints all even numbers from 1 to `N`.

Exercise 14: Check File Permissions

Write a bash script that:

- Prompts the user to enter a filename.
- Checks and displays whether the file is readable, writable, or executable.

Exercise 15: Monitor Memory Usage

Write a bash script that:

- Displays the total, used, and free memory using the `free -h` command.

Exercise 16: Display Logged-in Users

Write a bash script that:

- Lists all users currently logged into the system using the `who` command.

Exercise 17: Countdown Timer

Write a bash script that:

- Prompts the user to enter a number (seconds).
- Displays a countdown from that number to zero.
- Prints "Time's up!" at the end.

Exercise 18: Display File Line Count

Write a bash script that:

- Prompts the user to enter a filename.
- Counts and displays how many lines are in the file using `wc -l`.

Exercise 19: Simple Password Check

Write a bash script that:

- Prompts the user to enter a password.
- Checks if it matches a predefined password.
- Displays "Access granted" or "Access denied."

Exercise 20: Create a Directory Structure

Write a bash script that:

- Prompts the user to enter a main directory name.
- Creates the main directory with three subdirectories: `docs`, `images`, and `scripts`.
- Displays "Directory structure created successfully."